

DisasterSociety: Population-Calibrated Persistent Agents for Cross-Hazard Protective-Action Simulation

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Abstract—Post-disaster surveys usually retain whether and when households take protective action, but observe the preceding sequence of warnings, appraisals, preparation, coordination, and execution only partially and heterogeneously. Statistical models can estimate outcome and time distributions without generating this process; large language model (LLM) agents can generate detailed trajectories, but plausible narratives alone do not guarantee population calibration, temporal legality, or bounded information access. We formulate disaster protective-action simulation as population-calibrated causal event-history generation and present DisasterSociety, a generative multi-agent framework that separates calibration from process generation. Hazard and dataset adapters map event-specific evidence and survey fields into shared contracts. A persistent household agent updates cognitive and coordination states structured by the Protective Action Decision Model using only time-bounded, provenance-aware evidence, while a statistical event-history layer controls finite-window outcome and departure risk. A deterministic executor enforces legal state transitions and records complete evidence-to-action traces. The resulting research object is not respondent-level label prediction or narrative replay of a known disaster, but a joint distribution of auditable household histories. We define a multi-level evaluation protocol using Hurricane Irma and Carr Fire event packs that separately tests population and subgroup distributions, departure timing, causal and legal validity, retrospective Irma-to-Carr cross-hazard confirmation under frozen methods, and the functional effects of persistence, memory, feedback, and planning. This formulation makes cognitive modules falsifiable mechanisms and allows LLM-generated social processes to be assessed without treating classification accuracy as their primary measure.

Index Terms—Generative agents, multi-agent simulation, disaster evacuation, protective action, cognitive modeling, event history, calibration, cross-hazard validation.

I. INTRODUCTION

HOUSEHOLD protective action during a wildfire or hurricane develops over time. A household receives partial information, judges the source, assesses personal threat, checks transport and dependent-care constraints, prepares, and then departs or remains in place. A final evacuation label omits this history. Departure time alone also omits why a household became ready at that point.

Statistical choice and event-history models provide calibrated estimates of evacuation probability and departure time. They remain the appropriate reference when the task is supervised prediction. Their compact state, however, does not

produce a causal sequence of observations, appraisals, preparation actions, execution results, and later revisions. Rule-based agent-based models (ABMs) add temporal state and interaction [1], [2], but their behavioral transitions must be explicitly specified by the modeler.

Large language models (LLMs) offer a different kind of process model. LLM-based generative agents can retain memories, synthesize reflections, and plan behavior across multi-step interactions [3]. Platforms such as GenSim and AgentSociety extend this approach to large populations and reusable social environments [4], [5]. Disaster applications have also begun to combine behavioral theory with LLM reasoning; FLARE, for example, predicts wildfire evacuation decisions from post-event survey records [6]. Yet generative detail does not resolve empirical validity by itself. High-stakes crowd simulation therefore requires explicit evaluation of whether language-agent behavior aligns with observed human behavior [7]; process fidelity depends on variability and adaptation rather than a preferred action alone [8]; and an agent can appear more capable when it is given an unrealistically omniscient view of the interaction [9]. A fluent trajectory is therefore not sufficient evidence that a simulated population or decision process is valid.

In this work, we study *population-calibrated causal event-history generation* for disaster protective action. Given a time-stamped exogenous event stream and pre-decision household attributes, the task is to generate a joint population of histories, realized protective-action outcomes, and event times. A valid generator must satisfy three requirements. First, its aggregate and conditional outcome and timing distributions must remain anchored to observed data. Second, each household history must respect information availability and legal action ordering. Third, every simulated outcome must be traceable to delivered evidence, evolving appraisal, preparation or coordination, and execution. This task is neither respondent-level label prediction nor narrative reconstruction of a known disaster. Hurricane Irma and the Carr Fire serve as validation event packs for a reusable framework, not as the scope of the framework itself.

Three challenges follow from this formulation. The first is to connect population constraints to generated histories without preassigning a hidden binary outcome or exposing fitted probabilities to the agent. The second is to preserve causal state across time: environmental truth, household observation, and household belief must remain distinct, and no proposed action may bypass its prerequisites. The third is to separate reusable behavioral contracts from event-specific base rates,

timing, and evidence semantics while testing each cognitive component against the function it is intended to provide. In particular, memory, execution feedback, and planning cannot be treated as contributions merely because they are present in the architecture.

DisasterSociety assigns these responsibilities to separate system components. Dataset and hazard adapters map heterogeneous surveys and event records to a shared household, evidence, and action schema. A training-fitted statistical layer supplies continuous household propensity and discrete-time event risk without revealing either quantity to the agent. A persistent agent constrained by the Protective Action Decision Model (PADM) updates appraisal, preparation, coordination, and readiness using only evidence delivered by the current simulation time. A deterministic executor then enforces action legality, realizes permitted transitions, and writes the result back to the next agent state. Outcomes and event times are read from the executed history rather than inserted into the prompt or forced at the end of the simulation. Complete logs expose disagreements among calibrated risk, agent readiness, and realized action for later diagnosis.

We make three contributions.

- 1) We formulate disaster protective-action simulation as population-calibrated causal event-history generation, with population distribution, event timing, and process validity defined as separate empirical objects (Section III).
- 2) We develop a calibration-generation architecture that combines event-pack adapters, bounded perception, persistent PADM appraisal, statistical event risk, a legal executor, and complete causal logs (Section IV). Optional cognitive and social components are exposed as falsifiable mechanisms rather than assumed improvements.
- 3) We define a multi-level evaluation protocol using wildfire and hurricane event packs, statistical and agent-based baselines, paired mechanism ablations, and separate distributional, temporal, process, and cross-hazard criteria (Section V). Individual classification is retained only as a secondary diagnostic.

II. RELATED WORK

Our problem lies at the intersection of protective-action theory, sequential agent simulation, and empirical model validation. We organize prior work by the empirical object it explains: a final choice, a decision trajectory, or a population distribution of trajectories.

A. Protective Action, Evacuation, and Event Histories

The Protective Action Decision Model (PADM) separates environmental and social cues, predecision processes, threat, protective-action, and stakeholder perceptions, and situational facilitators and impediments [10]. This decomposition is important because a household can receive evidence without believing it, believe a threat without having a feasible action, or become ready without departing immediately. Statistical choice models summarize the resulting stay-or-go outcome, whereas survival and event-history models represent the timing

of an event conditional on remaining at risk [11]. Wildfire studies have consequently examined both evacuation choice and departure timing [12]. Reviews of wildfire evacuation simulation further identify heterogeneous response, behavioral-data scarcity, and the coupling of behavioral and traffic models as persistent challenges [13].

These traditions provide the behavioral variables and population-level targets needed for validation; they are not merely baselines to be displaced by an LLM. Their usual empirical outputs, however, are choices, hazards, or aggregate traffic patterns rather than auditable evidence-to-appraisal-to-action histories. DisasterSociety therefore retains statistical outcome and timing models as constraints while using a generative policy only for the intermediate, temporally ordered process.

B. Persistent Generative Agents and Social Simulation

Generative Agents introduced a memory-reflection-planning architecture for long-horizon social behavior [3]. Subsequent platforms have emphasized reusable environments and simulation infrastructure. GenSim supports customized scenarios, large populations, and long-run error correction [4]; AgentSociety combines LLM-driven agents, realistic urban environments, and computational social experiments [5]. The broader literature now spans cyber, physical, social, and hybrid systems, while continuing to identify perception, human alignment, action generation, and evaluation as open problems [14].

This line of work establishes that persistent state and environmental interaction can produce coherent multi-step behavior. It does not imply that memory, feedback, or planning automatically improves empirical validity. For disaster simulation, each component must instead be connected to an observable function—for example, recall continuity, response to execution failure, or consistency of plan revision—and tested at the trajectory level. Our focus is consequently not a more general agent platform, but a controlled way to turn persistent agents into population-calibrated event-history generators.

C. Behavioral Alignment and Multi-Level Validation

Recent work makes the behavior-realism problem explicit. Persona-Environment Behavioral Alignment formulates high-stakes crowd simulation as distribution matching and iteratively refines personas against expert behavioral benchmarks [7]. Process-oriented evaluation further shows that matching a preferred action is insufficient when agents fail to reproduce human variability and adaptation across rounds [8]. Information access is another validity boundary: LLM agents evaluated in omniscient settings can appear more successful than the same agents operating under realistic information asymmetry [9].

These concerns parallel established ABM validation practice. Pattern-oriented calibration compares a model against multiple weak patterns; in a protective-behavior case study, fitting both the magnitude and timing of adoption exposed structural flaws hidden by a single error criterion [15]. VAL-FRAM likewise validates generated schedules along temporal,

spatial, and structural dimensions rather than reducing an activity history to one endpoint [16]. Taken together, this work argues against using either individual accuracy or narrative plausibility as a sufficient statistic. DisasterSociety follows the multi-level principle but couples three objects that are commonly evaluated separately: the population outcome distribution, the departure-time distribution, and the legality and causal coherence of each generated history.

D. LLMs for Disaster and Emergency Behavior

Disaster-specific LLM research is moving from unconstrained role play toward behavioral structure. FLARE integrates behavioral theory, chain-of-thought reasoning, and memory-based reinforcement learning to predict individual wildfire evacuation decisions from post-event survey records [6]. Hierarchical Generative Agents model evacuation as sequential high-level goal setting, route reasoning, and low-level navigation, with calibration to empirical evacuation data [17]. PAVE addresses a complementary emergency problem: it structures how agents perceive, assess, delimit, and recover from legitimate rule violations, and evaluates properties such as authority deference and bounded scope [18].

The closest work thus offers, respectively, theory-informed outcome prediction, cognitively structured evacuation and navigation, or property-based emergency reasoning. DisasterSociety changes the primary empirical object to a joint population of household event histories. A separate calibration layer constrains finite-window outcomes and timing; the persistent agent generates only causally available observation–appraisal–action transitions; and the executor makes every transition auditable. This calibration–generation separation is the methodological distinction on which the remainder of the paper builds.

III. PROBLEM FORMULATION

Let the simulation window contain ticks $t = 0, \dots, T - 1$ and households $\mathcal{V} = \{1, \dots, N\}$. The exogenous event stream is

$$\mathcal{E}_{0:T} = \{e_k\}_{k=1}^K, \quad (1)$$

where each event contains a publication time, type, geographic scope, source, provenance, and uncertainty. Household i has pre-decision attributes X_i , including household composition, mobility constraints, vehicle access, previous experience, information access, and coarse location.

The simulator generates

$$\{\mathcal{H}_i, Y_i, \tau_i\}_{i=1}^N, \quad (2)$$

where $\mathcal{H}_i$ is an evidence-to-action history, Y_i is the finite-window protective-action outcome, and τ_i is the simulated departure time if departure occurs. An explicit stayer is a complete finite-window non-event. Only an incomplete survey history is right-censored.

Environmental truth, household observation, and household belief remain distinct:

$$W_t \neq O_{i,t} \neq B_{i,t}. \quad (3)$$

An event can enter $O_{i,t}$ only if it has been published by t , is relevant to the household, and passes the configured delivery

rule. The policy cannot observe the survey outcome, observed departure time, fitted propensity, random draw, or future event.

The empirical objective has three parts:

$$\mathcal{L}_{outcome} = d(\hat{P}(Y), P_{survey}(Y)), \quad (4)$$

$$\mathcal{L}_{time} = d(\hat{P}(\tau), P_{survey}(\tau)), \quad (5)$$

$$\mathcal{L}_{process} = \phi(\{\mathcal{H}_i\}), \quad (6)$$

where d denotes distributional or calibration error and ϕ contains legality, temporal-causality, and mechanism diagnostics. No single classification metric is used as evidence for all three parts.

IV. THE DISASTERSOCIETY FRAMEWORK

A. System Boundary and Adapters

Fig. 1 shows the framework boundary. A dataset adapter maps heterogeneous survey variables to a common household and outcome schema. A hazard adapter maps official orders and public hazard records to typed, timestamped evidence. The universal simulator reads only the common schema. Outcome fields remain in the evaluation store and cannot enter the agent observation.

B. Training-Fitted Household Propensity

A regularized contextual logistic model is fitted on the source training partition:

$$p_i = P(Y_i = 1 \mid X_i) = \sigma(\beta_0 + \beta^\top X_i). \quad (7)$$

The features cover household composition, home-area context, mobility and transport, information access, age, and decision role. Missing values are imputed within the training pipeline. The resulting p_i is continuous. It is never converted into a fixed binary eligibility class and is hidden from the agent.

For target-training-only robustness analysis, we preserve the source-trained slopes and ranking and fit one intercept on the target training split:

$$p_i^{(a)} = \sigma(\text{logit}(p_i) + a). \quad (8)$$

The Carr test split is excluded when estimating a . This adapter measures whether a small hazard-specific base-rate parameter can repair cross-event calibration. It is not reported as zero-shot transfer.

C. Training-Fitted Timing Mass

Let $w_t \geq 0$ be the fraction of observed training departures in tick t , with Jeffreys smoothing:

$$w_t = \frac{n_t + 1/2}{\sum_{s=0}^{T-1} n_s + T/2}, \quad \sum_{t=0}^{T-1} w_t = 1. \quad (9)$$

The Irma-trained weights define the frozen cross-hazard timing condition. A Carr-training-only timing adapter replaces $\{w_t\}$ while keeping the agent core and household propensity slopes fixed. We also test Gaussian smoothing over timing bins. Five-fold likelihood selection on Carr training data chooses zero smoothing, so the smoothed variant is rejected before test interpretation.

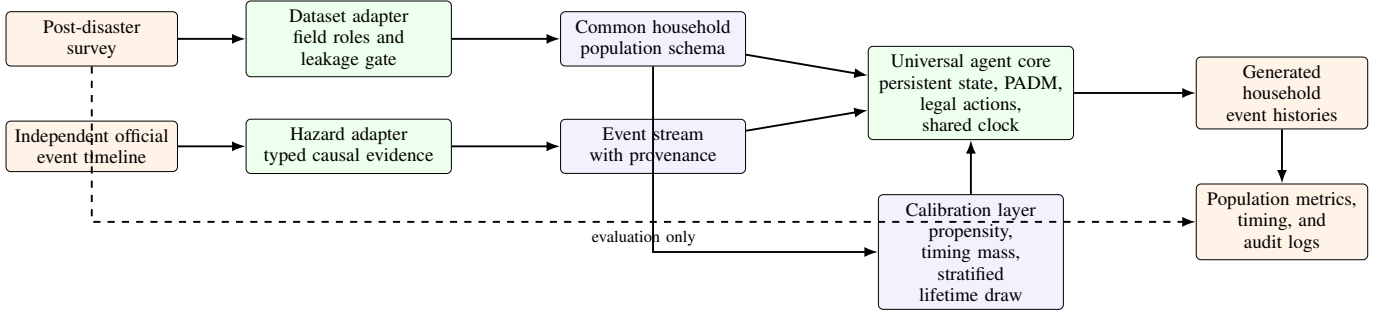


Fig. 1. System boundary. Survey outcomes are separated from agent inputs. Hazard and dataset adapters feed a shared agent core; the calibration layer controls outcome and timing mass without exposing hidden targets to agents.

D. Legal-Risk-Set Deferred Mass

The main controller consumes timing mass only when the household is in `PREPARING` or `COORDINATING`. Let $L_{i,t} \in \{0, 1\}$ indicate membership in this legal pre-departure risk set. Let $d_{i,t}^-$ be timing mass deferred before tick t and $c_{i,t}^-$ be mass already consumed. The effective mass is

$$q_{i,t} = \begin{cases} \min(1 - c_{i,t}^-, w_t + d_{i,t}^-), & L_{i,t} = 1, \\ 0, & L_{i,t} = 0. \end{cases} \quad (10)$$

The update is

$$d_{i,t}^+ = \begin{cases} 0, & L_{i,t} = 1, \\ \min(1 - c_{i,t}^-, d_{i,t}^- + w_t), & L_{i,t} = 0, \end{cases} \quad (11)$$

$$c_{i,t}^+ = \min(1, c_{i,t}^- + q_{i,t}). \quad (12)$$

The corresponding conditional hazard, retained in the audit log, is

$$h_{i,t} = \frac{p_i q_{i,t}}{1 - p_i c_{i,t}^-}. \quad (13)$$

This construction does not create an invisible pending departure. A household that is not ready accumulates timing mass; the mass becomes available at the next legal tick. A household that never reaches the risk set remains a process non-event, exposing disagreement between the calibrated outcome head and the generated preparation process.

E. Variance-Controlled Lifetime Realization

Each household receives one lifetime uniform draw. For $K = 5$ registered seeds, a stable household hash supplies an offset $o_i \in \{0, \dots, K - 1\}$ and jitter $j_i \in [0, 1)$. Run s uses

$$u_{i,s} = \frac{(s + o_i) \bmod K + j_i}{K}. \quad (14)$$

The five runs therefore place each household once in every uniform stratum, as in Latin-hypercube sampling [19]. Departure occurs at the first legal tick satisfying

$$u_{i,s} < p_i c_{i,t}^+. \quad (15)$$

This reduces five-seed Monte Carlo noise while preserving the marginal lifetime probability p_i . The same hash and registered seeds are used in all paired comparisons.

F. Persistent Agent State and PADM Appraisal

At tick t , agent i maintains

$$S_{i,t} = (X_i, D_{i,t}, B_{i,t}, M_{i,t}, P_{i,t}), \quad (16)$$

where $D_{i,t}$ is objective dynamic state, $B_{i,t}$ is structured PADM appraisal, $M_{i,t}$ is optional structured memory, and $P_{i,t}$ is an optional short plan. The main evaluation profile persists the legal stage and PADM state. A separate mechanism ablation activates execution feedback, memory, and plan revision.

The observation includes only evidence delivered by the current tick. Each item retains time, source, credibility, geographic relevance, and provenance. The policy updates threat appraisal, protective-action feasibility and cost, source trust, contradictions, and a readiness value in `{not_ready, considering, ready}`. It returns one JSON object containing a legal action, intended stage, readiness, rationale, cited evidence identifiers, a plan of at most three steps, and optional messages. Citations not present in the current observation are removed.

G. Action State Machine and Execution

The action state machine includes

$$\begin{aligned} \text{UNAWARE} &\rightarrow \text{MONITORING} \\ &\rightarrow \text{PREPARING} \leftrightarrow \text{COORDINATING} \\ &\rightarrow \text{DEPARTING} \rightarrow \text{EVACUATED}, \end{aligned} \quad (17)$$

with legal finite-window branches to `STAYING`. A policy may declare readiness, but it cannot force departure while the controller is active. The executor checks action-stage compatibility, commits the legal transition, and records acceptance, rejection, objective updates, and completion.

H. Synchronous Concurrent Kernel

All households share one simulation clock. At each tick the engine:

- 1) replays exogenous events and delivers due messages;
- 2) freezes a read-only world snapshot;
- 3) constructs bounded observations;
- 4) runs household decisions concurrently;
- 5) waits at a barrier before any decision can affect another;
- 6) applies the departure controller and validates actions;
- 7) commits transitions and queues later messages;

TABLE I
EVENT PACKS AND FORMAL EVALUATION COHORTS

	Irma 2017	Carr Fire 2018
Valid raw responses	1,263	647
Completed questionnaires	921	335
Valid stay/go outcome	917	330
Complete departure time	539	292
Formal test households	87	68
Formal test evacuated	42	60
Formal test stayed	45	8
External event source	HEvOD	CAL FIRE

8) writes the causal trajectory and validity audit.

The experiments use up to 32 concurrent model calls. Concurrence changes wall time, not simulated time. A message created at tick t cannot affect another agent inside that tick.

I. Audit and Validity Gates

Every transition stores the observation, cognition before and after, retrieved memories, proposed action, controlled action, execution result, and departure control record. Every model call stores the prompt hash, seed, status, token counts, latency, and cache status. Outputs are validated against a typed schema. A malformed response receives one repair attempt; a failed request falls back to a rule policy and is recorded. A run is invalid if fallback exceeds 1%, an output is missing, a future event enters an observation, or an illegal transition exceeds the registered tolerance.

V. EXPERIMENTAL DESIGN

A. Data and Evaluation Cohorts

Table I summarizes the two event packs. The Irma survey contains 1,263 valid responses and 921 completed questionnaires. Official evacuation orders come from the Hurricane Evacuation Order Database. The formal within-event test uses the Brevard County public-signal cohort associated with HEvOD event 21: 87 households, including 42 evacuees and 45 stayers.

The Carr survey contains 330 valid stay-or-go responses among completed records, with 293 evacuees and 37 stayers, and 292 complete departure times. The retrospective Carr test contains 68 households, including 60 evacuees and eight stayers. Carr-train adapter experiments use 197 training households; the test split remains evaluation-only. Independent CAL FIRE records provide the public event timeline.

The surveys are convenience samples. We report survey-conditioned reconstruction, not county or state population estimates. Exact respondent locations, observed social ties, respondent GPS trajectories, and link-level traffic measurements are unavailable.

B. Splits, Leakage Control, and Freezing

Each event pack uses a household-stratified split with seed 42. All records derived from one respondent remain in the same partition. Survey outcomes, departure times, and post-decision fields are tagged as targets or diagnostics and cannot

enter the policy prompt. The Irma training partition fits the contextual propensity and frozen timing weights. Development experiments select the calibrated deferred-mass controller and state-only profile before the Irma test summary is generated.

Carr is not described as a pristine external validation because the project had inspected it before the final protocol. We call it retrospective cross-hazard confirmation after method freezing. The frozen condition uses Irma propensity and timing components without Carr labels. Separate target-training-only robustness conditions fit an intercept and timing weights on Carr train and never use Carr test labels.

C. Baselines and Ablations

The within-Irma comparison includes:

- 1) a regularized logistic outcome model;
- 2) a contextual propensity model using harmonized household features;
- 3) a non-LLM PADM rule ABM;
- 4) a one-shot stateless LLM;
- 5) the calibrated persistent process.

All LLM experiments use `deepseek-v4-flash` through an OpenAI-compatible interface, temperature 0.2, structured JSON output, and registered seeds 41–45. We report model-call counts and output validity for computational reproducibility.

The mechanism ablation uses a frozen 30-household Irma subset and four cumulative profiles: stateless, state-only, state plus execution feedback, and full-persistent with structured memory and plan revision. Each profile uses five paired seeds. This design tests whether named modules change empirical outcomes; architectural availability alone is not counted as evidence.

D. Metrics and Uncertainty

Outcome metrics are Macro-F1, balanced accuracy, Brier score, expected calibration error (ECE), and absolute population-share error. We report individual discrimination as a diagnostic because the primary object is the population event-history distribution. Five-seed ensemble probabilities equal the household's realized action frequency across seeds.

Departure-time metrics are Wasserstein-1 distance in hours, integrated absolute difference between empirical cumulative distribution functions (CDF IAE), and absolute mean-time error. Process metrics include illegal-transition rate, future-evidence count, fallback rate, PADM evidence response, source-appraisal coverage, readiness-to-execution diagnostics, repeated actions, feedback response, and plan revision.

Household-level effects use 10,000 paired bootstrap samples. Timing-adapter effects use paired seed statistics. We report point effects and 95% percentile intervals. An interval crossing zero is treated as inconclusive, regardless of the direction of the point estimate.

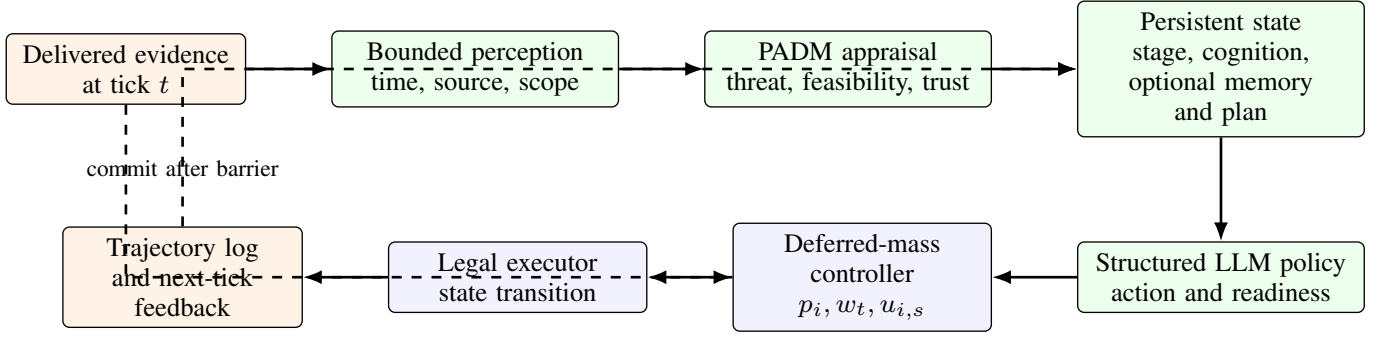


Fig. 2. Household decision and execution loop. The agent controls cognition, preparation, and readiness. The calibration layer controls event realization, and the executor enforces legal transitions.

TABLE II
EMPIRICAL EVALUATION COMPONENTS

Component	Empirical purpose
Data validity	Schema, provenance, split, and leakage checks
Irma reconstruction	Within-event outcome and timing evaluation
Mechanism ablation	Persistent state, feedback, memory, and planning
Frozen transfer	Retrospective Irma-to-Carr confirmation
Target-train robustness	Carr intercept, timing, and resolution adapters
Process validity	Sequence legality and causal-information checks

E. Evaluation Structure

VI. RESULTS

A. Irma Within-Event Reconstruction

Table III reports the 87-household Irma test. The calibrated persistent ensemble has the highest Macro-F1 (0.551), balanced accuracy (0.553), and lowest population-share error (0.018). The contextual propensity has a lower Brier score (0.261 versus 0.269), while the persistent ensemble has lower ECE (0.124 versus 0.154).

Relative to the contextual propensity, the persistent-process Macro-F1 effect is $+0.058$ with a 95% household-bootstrap interval of $[-0.023, 0.138]$. The point estimate favors the persistent process, but the interval does not establish a classification improvement. The propensity-minus-process Brier effect is -0.007 with interval $[-0.031, 0.015]$, which indicates a small point disadvantage for the process model. The share-error improvement is 0.027 with interval $[-0.041, 0.047]$. The persistent process therefore improves the observed group-share and ECE point estimates, but the sample does not support a broad claim of statistical superiority.

Across seeds, the calibrated persistent process realizes an evacuation share of 0.501 ± 0.052 against the observed 0.483. Its seed-level Macro-F1 is 0.525 ± 0.025 . All five runs are valid, with no fallback, schema repair, missing output, leakage, or invalid transition.

B. Persistence Is the Identifiable Agent Mechanism

Table IV reports five-seed ensemble results for the paired 30-household mechanism-ablation subset. Stateless agents never maintain the legal preparation stage across ticks, so their ensemble generates no departures and obtains a Macro-F1 of 0.362. State-only raises Macro-F1 to 0.700 and reduces Brier score from 0.433 to 0.203.

The household-paired Macro-F1 effect for state-only versus stateless is $+0.331$, with a 95% interval of $[0.125, 0.522]$. Brier improvement is $+0.230$ with interval $[0.039, 0.423]$. Both intervals exclude zero. State+feedback versus state-only has a Macro-F1 effect of -0.035 with interval $[-0.116, 0.000]$. Full-persistent versus state-only has an effect of -0.002 with interval $[-0.132, 0.130]$. The tested data support persistent state. They do not show an additional predictive gain from feedback, memory, or plan revision. Full-persistent obtains a lower share-error point estimate, but its paired improvement over state-only is inconclusive.

C. Frozen Irma-to-Carr Confirmation

The Carr test has a much higher observed evacuation share than the Irma cohort: 0.882 versus 0.483. Table V shows that the frozen Irma contextual propensity retains some household ranking on Carr, with a Macro-F1 of 0.584 and balanced accuracy of 0.679, but predicts a share of only 0.584. The frozen persistent-process ensemble has a Macro-F1 of 0.595 and a share of 0.597. Its Macro-F1 effect over the contextual propensity is $+0.012$ with a 95% interval of $[-0.087, 0.092]$. Its Brier improvement is 0.004 with interval $[-0.013, 0.022]$. Both comparisons are inconclusive.

The transfer result separates ranking from base-rate calibration. The frozen model ranks some Carr households correctly, but it underestimates the event share by 0.285. The Carr-training-only intercept adapter shifts the process ensemble share to 0.885, within 0.0029 of the observed 0.882, and lowers Brier score to 0.103. Its Macro-F1 falls to the majority-class value because the high fitted base rate moves nearly all households above the 0.5 threshold. This is a calibration repair, not an individual-classification improvement.

D. Departure Timing and Adapter Resolution

Table VI reports Carr timing experiments. Frozen Irma timing transfers poorly: Wasserstein distance is 27.41 hours

TABLE III
IRMA TEST RESULTS ON THE FROZEN 87-HOUSEHOLD COHORT

Method	Runs	Macro-F1	Bal. Acc.	Brier	ECE	Share error
Logistic outcome model	1	0.490	0.539	0.248	–	0.096
Contextual propensity	1	0.494	0.496	0.261	0.154	0.045
PADM rule ABM	5	0.326	0.500	0.517	–	0.517
One-shot stateless LLM	5	0.421	0.507	0.312	–	0.191
DisasterSociety calibrated persistent	5	0.551	0.553	0.269	0.124	0.018

TABLE IV
FIVE-SEED PERSISTENT-AGENT MECHANISM ABLATION

Profile	Macro-F1	BAcc	Brier	Share err.
Stateless	0.362	0.500	0.433	0.433
State-only	0.700	0.717	0.203	0.053
State+feedback	0.665	0.688	0.208	0.080
Full-persistent	0.697	0.699	0.224	0.027

and CDF IAE is 0.256. Replacing only timing weights with Carr-training estimates reduces Wasserstein distance to 8.00 hours without changing the outcome ensemble. The joint intercept and timing adapter yields 7.40 hours. Keeping the same fitted components and changing resolution from 12 hours to six hours reduces Wasserstein distance to 5.49 hours and CDF IAE to 0.0446. The paired Wasserstein improvement from higher temporal resolution is 1.91 hours, with a 95% seed bootstrap interval of [1.11, 2.69].

The six-hour condition doubles transitions from 3,060 to 6,120 and increases recorded policy decisions from 2,376 to 4,474. Outcome metrics remain identical to the 12-hour joint-adaptation condition. Training-only cross-validation selects zero Gaussian smoothing, so smoothing is not included as a reported experimental condition. The 12-hour configuration remains the default probability-calibrated process; the six-hour condition is used only for high-resolution timing analysis.

E. Sequence Legality and Process Diagnostics

The calibrated Irma process generates no illegal execution in any of its five main runs. About 68.4% of tick-level controller evaluations occur outside the legal pre-departure risk set, so the corresponding timing mass is deferred. Among agent-ready decisions in the risk set, the mean hazard-not-triggered rate is 0.891. This value shows that readiness and event realization are separate quantities. The PADM evidence-response rate and source-appraisal coverage are both 1.0 in the main Irma profile; the repeated-action rate is 0.412.

Irma departure-time reconstruction remains weak. Of 42 observed departures with timing, ten occur before the selected public signal and are excluded from post-signal timing attribution. For the remaining departures, the calibrated 12-hour process has a Wasserstein distance of 17.69 hours, CDF IAE of 0.231, and mean-time error of 16.52 hours. The trajectory is legally executable, but the selected Irma signal and coarse timing model do not reproduce observed departure timing accurately.

The five frozen Carr transfer runs contain 3,060 transitions, zero temporal leaks, zero rejected actions, and zero fallbacks. The 12-hour joint-adaptation condition also has 3,060 legal transitions and no leak or fallback. The six-hour condition contains 6,120 legal transitions and no leak or fallback. These checks establish simulator validity, not behavioral fidelity.

VII. DISCUSSION

A. What the Results Support

The mechanism ablation provides the clearest mechanism evidence. A stateless policy cannot sustain preparation across ticks and fails to enter the legal departure risk set. Persisting stage and appraisal state raises ensemble Macro-F1 by 0.331, and the bootstrap interval excludes zero. The effect measures the value of continuity inside this simulator, not a universal advantage for every LLM-agent task.

The Irma main experiment provides a different result. The calibrated persistent process matches the observed evacuation share more closely and has lower ECE than the contextual propensity, but its Brier score is slightly worse and its Macro-F1 interval crosses zero. Population calibration and individual classification are not interchangeable. The process layer can generate a more useful group realization without becoming a better supervised classifier.

The Carr experiments expose the limits of cross-hazard transfer. Irma-trained household slopes preserve some discrimination, while the Carr event has a different base rate and departure-time distribution. One target-training-only intercept and a timing histogram repair these quantities without changing the universal agent state or action contracts. The reusable object is therefore a shared cognitive and execution core plus explicit hazard and event adapters, not one fixed probability distribution for every disaster.

B. Why More Cognitive Modules Did Not Improve Prediction

The full-persistent profile includes execution feedback, structured memory, and plan revision, but it does not outperform state-only. Three explanations remain plausible. First, the short 72-hour Irma window may not require long-memory retrieval. Second, the available public signal is sparse, so a richer memory cannot add unavailable evidence. Third, the outcome controller already fixes much of the finite-window distribution, leaving the cognitive modules to change sequence quality rather than the final label. The present quantitative data do not identify such a sequence-quality effect.

We therefore do not list memory, feedback, or plan revision as independently validated contributions. They remain optional

TABLE V
CARR OUTCOME RESULTS: FROZEN TRANSFER AND TARGET-TRAIN ROBUSTNESS

Condition	Fit data	Macro-F1	BAcc	Brier	ECE	Share error
Always evacuate	none	0.469	0.500	0.118	0.118	0.118
Carr contextual reference	Carr train	0.469	0.500	0.116	0.106	0.021
Frozen Irma contextual	Irma train	0.584	0.679	0.201	0.321	0.299
Frozen persistent process	Irma train	0.595	0.650	0.197	0.285	0.285
Carr intercept+timing adapter	Irma+C. train	0.469	0.500	0.103	0.038	0.0029

TABLE VI
CARR DEPARTURE-TIME RECONSTRUCTION

Condition	Timing fit	Tick	Wasserstein (h)	CDF IAE	Mean error (h)
Frozen transfer	Irma train	12 h	27.41	0.2561	25.10
Timing-only adaptation	Carr train	12 h	8.00	0.0676	3.43
Intercept+timing adaptation	Carr train	12 h	7.40	0.0628	2.56
Intercept+timing, 6 h tick	Carr train	6 h	5.49	0.0446	2.73

mechanisms with typed inputs, outputs, and direct diagnostics. This negative result is useful because it prevents architectural complexity from being mistaken for empirical support.

C. Implications for Generative Social Simulation

The framework treats an LLM as a constrained process policy. The LLM interprets evidence and proposes legal actions; it does not control environmental truth, population base rate, or random event realization. This boundary addresses a common validation failure in generative social simulation: plausible individual text can coexist with a wrong aggregate distribution.

The deferred-mass interface also keeps model disagreement observable. If an agent never prepares, its statistical propensity does not silently force an end-window departure. If an agent prepares but its lifetime draw is above the consumed mass, readiness does not force an event. These cases remain in the trajectory log and can be examined as process-friction diagnostics.

VIII. LIMITATIONS AND ETHICAL SCOPE

Both surveys are convenience samples and contain self-reported behavior. The formal cohorts are small, especially the 30-household ablation and the eight-stayer Carr test minority. Bootstrap intervals reflect sampling within these cohorts and do not convert them into representative population samples.

Carr was inspected during earlier project development. The cross-hazard result is retrospective confirmation after method freezing, not a pristine prospective validation. Carr-training adapters use target training labels and must be interpreted as robustness analyses. They cannot be described as zero-shot generalization.

The event timelines are coarse. The Irma timing analysis begins at one public HEVOD signal, although 23.8% of observed timed departures in the cohort occur before that signal. Carr timing improves after a hazard-specific adapter, which confirms that departure clocks are event dependent.

The simulator does not recover real social ties. Social-graph messaging is an optional mechanism and is disabled in

the calibrated evaluation profile. The study does not model physically accurate wildfire spread, hurricane wind, traffic flow, road capacity, route optimality, or clearance time because the available surveys do not contain respondent GPS paths or link-level traffic ground truth.

The LLM is accessed through a forwarding service whose model-specific price and immutable snapshot identifier are not independently verified. Prompts, outputs, seeds, hashes, and cache records are retained, but provider-side reproducibility remains a limitation. Language-model rationales are simulator audit records. They are not evidence of real human cognition.

Synthetic-agent results should not be used to issue evacuation orders, allocate emergency aid, or infer the behavior of a named respondent. The system is designed for method research and scenario analysis under explicit uncertainty. Human decision makers remain responsible for any operational interpretation.

IX. CONCLUSION

DisasterSociety combines a training-fitted population model with persistent, evidence-bounded household agents and a legal event-history executor. The deferred-mass controller preserves continuous household propensity without preassigning an evacuation class or forcing departure at the end of the window. Household-offset Latin-hypercube draws reduce five-seed realization noise, while the shared-clock kernel makes future-information and illegal state transitions auditable.

Experiments on Hurricane Irma and the Carr Fire support persistent state as the main agent mechanism. Irma group-share and ECE point estimates improve, but individual classification superiority over the contextual propensity is not established. Frozen Irma-to-Carr transfer retains partial discrimination and misses the Carr base rate and timing distribution. Small Carr-training-only adapters repair calibration and timing, showing that a cross-hazard simulator needs a shared process core and explicit event-level parameters. Memory, feedback, and plan revision do not add identifiable predictive value in the current data and remain unvalidated beyond their process diagnostics.

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